PROJECT TECHNICAL DOCUMENTATION

BioSpecInfo

Web platform for computational chemistry, physics and mathematics

Verification & Validation (V&V) & Benchmark Report

DOCUMENT CONTROL

Identifier	`BSI-DOC-02-EN`
Title	Verification & Validation (V&V) & Benchmark Report
Software version	**bsi-v250**
Revision (commit)	`2c08fbc`
Date of issue	2026-11-12
Author	Samuele Pio Provenzano
Language	English
Status	Issued with version bsi-v250

Document generated from the versioned source in the project repository. The version and revision shown here and in the footer of every page identify exactly the state of the software described.

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Verification & Validation (V&V) & Benchmark Report — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Purpose	Document the verification method, validation data for the spectral predictors, cross-device performance and declared model limitations.

1. Scope and V&V strategy

- **Verification:** the software is built correctly (syntax, no runtime errors, expected component behaviour).
- **Validation:** the scientific results (spectral predictors, descriptors) are compared against reference experimental databases.

2. Software verification (QA)

Technique	Description	Status
Syntax checking	`node --check` on the critical JavaScript blocks on every change.	✅ In use
End-to-end (E2E) testing	Automation with **Playwright** on headless Chromium: open sections, draw/load molecules, verify panels (notations, spectra, functional groups, ADMET, similarity), 2D/3D animations. Counting of console/page JavaScript errors (acceptance criterion: **0 non-network errors**).	✅ In use
HTTP-server verification	RDKit WASM requires an HTTP context (CORS): tests run on a local server to reproduce the real environment.	✅ In use
Manual cross-device testing	Testing on real devices (see §4).	✅ In use
Automated unit-test suite (Vitest/Jest)	—	🔜 Roadmap
Coverage ≥ 90%	—	🔜 Roadmap

> Every release is verified E2E with Playwright under a "zero JS errors" criterion before merging to the main branch.

3. Scientific validation of the spectral predictors

BioSpecInfo's spectral predictors use **heuristic group-additivity / structural rule** models (literature reference tables), chosen to guarantee **instant in-browser response** on any device.

3.1 ^{1}H-NMR benchmark vs SDBS

Metric	Reported value
Reference database	**SDBS** (Spectral Database for Organic Compounds, AIST)
Mean Absolute Error (MAE)	**0.31 ppm**
Signals within ±0.5 ppm	**89%**

3.2 IR benchmark (C=O band) vs NIST

Metric	Reported value
Reference database	**NIST Chemistry WebBook**
Mean Absolute Error (MAE)	**11 cm^{-1}**
Bands within ±30 cm^{-1}	**90%**

3.3 Physicochemical properties

Descriptors (molecular weight, formula, LogP, TPSA, HBD/HBA, rings, stereocentres) are computed with **RDKit MinimalLib** and comparable with **PubChem** and **DrugBank**. Reproducible verification example: *cholesterol → $C_{27}H_{46}O$, 4 rings, 8 stereocentres*.

> **Benchmark data source:** measurements carried out by the author as part of the undergraduate thesis. The values are to be understood as validation for **educational and qualitative functional-group identification** purposes, not as a substitute for reference quantum-mechanical calculations.

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4. Cross-device performance

Smoothness tests of the interactive renderers (3D viewers, animations, editor) across a range of devices:

Device	Result
iPhone / iPad	30–60 fps
Windows / macOS laptop	60 fps
Raspberry Pi 4	30–60 fps

Size of the main `index.html` file: **≈ 4.5 MB** (the full package, including WASM modules and 3D assets, is larger but remains entirely static and cacheable offline).

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5. Declared model limitations

In line with the principle of scientific transparency:

1. **Additivity model.** The NMR/IR predictors sum group contributions and do

not fully model extended chemical-environment effects (long conjugation, anisotropic effects, intramolecular hydrogen bonds). Larger deviations are expected e.g. for **strongly conjugated carbonyls** or edge cases such as **formic acid**.

1. **Qualitative/educational spectra.** The simulated spectra serve to recognise

functional groups and understand structure–spectrum relationships, not to replace certified experimental data.

1. **Network dependency for some functions.** IUPAC/common name, CAS number, GHS

pictograms and real 3D models come from PubChem: without network the app shows a fallback and remains fully functional on local data.

1. **Two-way accuracy.** For rigorous use, direct links to the official databases

(NIST, SDBS, MassBank, nmrshiftdb2) are integrated into the interface.

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6. Release acceptance criteria

- ✅ `node --check` passes without errors on the modified modules.
- ✅ Playwright E2E: 0 non-network JS errors.
- ✅ No visible regression in the touched sections.
- ✅ Service Worker cache version bumped.
- ✅ `tools/verifica-farmaci.js`: no defects; every deviation recorded with its reason.
- ✅ `verifica_guida`: whatever the documentation promises exists in the code.

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7. Verification battery — state at version `bsi-v250`

Verification is no longer only functional end-to-end testing: it includes benches dedicated to the properties a functional test does not observe (stability over time, network degradation, data accuracy).

7.1 Composition

Family	Benches	Subject
Scientific data	`verifica-farmaci`, `test_spettri`, `test_assi`, `test_assi_canvas`, `test_costanti`, `audit_dati`, `test_v237`, `test_3d238`	structure vs molecular weight, functional-group recognition, axis conventions, physical constants; the twenty astrophysics, physics and mathematics sections against known values or an independent second method; the 3D tabs of 29 sections (311 computation checks, every tab opened at 1280 and 390 px)
AI agent	`test_ko`, `test_404`, `test_503`, `test_firma`, `test_attesa`, `test_attesalunga`, `test_tetto`, `test_nucleo`, `browser_ko`, `browser_prova`	retired model, CORS, overload, reasoning signatures, token ceiling, long waits, memory of unreachable providers
Stability	`audit_stabilita`, `audit_promesse`, `audit_quota`, `test_filemanager`, `test_doppioinvio`, `test_sw`, `test_visore3d`	long sessions, rejected promises, exhausted storage, unavailable archive, overlapping submissions, degraded network, WebGL contexts
Interface	`browser_reset`, `browser_proxy`, `browser_proxyui`, `browser_rdkit`, `browser_lab`, `browser_frontiera`, `test_aggiorna`, `test_controllaagg`, `test_spectra238`, `test_lezione`, `test_spectra_studio`, `test_spectra_rig`, `test_spectra_figura`, `test_guidaproxy`	panels, proxy, RDKit laboratory, updates (including the «Check» button pressed in the three cases: up to date, new version, site unreachable); the Spectra Live console on phone and large screen, navigator, level and concentration, creator recognition, the face's eyes and sleep, a single Spectra button; the guided lesson in all 194 sections (chapters from the tabs, controls shown at work and put back as they were, check question, summary); Spectra Studio: the computations of the three lessons, the body driver on its bench's mannequin, the always-present embedded body (a packed block, not a `.glb`) and the real rigged body, the voice chain, the lessons on two screens and without WebGL; the automatic rig on a synthetic humanoid (skeleton, weights, hood with the face only, night blue); Spectra with the body in the four AI sections, the face-only icon, no substitute, the figure's reactions
Consistency	`verifica_guida`	documentation ⟷ code correspondence, conflict markers across 58 files

7.2 Properties verified that a functional test does not observe

Property	Method	Measured result
Absence of memory leaks	194 sections opened for 5 rounds, DOM node count per round	construction +26,876 nodes on the first round, **+0 on the four that follow**
Survival of exhausted storage	`localStorage.setItem` forced to throw	10 pages out of 10 stay operational

Property	Method	Measured result
Network degradation (≠ no network)	requests held for 20 s, interception at context level	answer served from cache in **3,507 ms** (threshold 3,500 ms)
Spectrum reproducibility	drawn twice, compared byte for byte	identical
WebGL contexts	viewer constructions counted across 10 molecules	from 7 down to **1**

7.3 Notes on method

A bench that measures nothing passes, and that is the most insidious way to obtain a false guarantee. While the battery was being built, several cases of vacuous measurement were found and corrected, among them:

- Playwright's `page.route` does **not** intercept requests originated by the

Service Worker: `context.route` is required. With the former, the "degraded network" test was measuring a perfectly healthy network and concluding in 16 ms.

- Probing `localStorage` exhaustion with a one-byte write proves nothing: after

saturation, one byte always finds room.

- The ticks of an axis drawn on `<canvas>` are not readable as text: they must

be collected by intercepting `fillText`, and the bottom-most line is the axis *title*, not a tick.

Hence the rule adopted: **every bench that simulates a condition must count how many times the simulation actually fired**, and fail if that count is zero.